

ASHNA ALI

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PROFESSIONAL SUMMARY

Entry-level Software Developer with experience in full-stack development and applied machine learning. Skilled in Python, JavaScript, and SQL, with hands-on expertise in PyTorch, TensorFlow, and React. Passionate about building scalable, data-driven solutions and eager to contribute while continuously expanding technical skills.

SKILLS

Languages: Python, Java, C++, C#, SQL, JavaScript, TypeScript, Ruby on Rails, NodeJS, ReactJS, Swing GUI

Databases: MS SQL Server (2022), MySQL, Postgres, MongoDB (NoSQL)

Tools: AWS (EC2, ECS, S3), MS Visual Studio (2022), Jenkins (CI/CD), JupyterLab, Scikit-Learn, IntelliJ, Eclipse, PyCharm, JIRA, MS Teams, MS Visual Studio Code, GitHub, Docker

EDUCATION

Master's of Science Computer Science (Data Science Emphasis)

December 2024

University of Missouri-Kansas City

GPA: 3.48

Master's Projects

AI Diabetes Health Coach App

Aug 2024-Dec 2024

- Developed gender-specific diabetes prediction models using XGBoost and deep learning, achieving 80–90% accuracy
- placed 3rd at University of Missouri–Kansas City Hack-A-Roo.
- Integrated large-language model (LLM) to generate personalized health recommendations from model outputs.
- **Tools:** Python, JupyterLab, PyTorch, TensorFlow, Ollama (open-source Llama 3.2 LLM)

CloudBuilder

Aug 2024-Dec 2024

- Designed and implemented automated CI/CD pipelines for ReactJS applications using AWS (S3, EC2) and Docker.
- Integrated containerization, monitoring, and reverse proxy to enable scalable and secure deployments with Next.js.
- **Tools:** AWS (S3, EC2), Docker, Next.js, ReactJS

IoT Mining Safety Monitoring System

Aug 2024-Dec 2024

- Developed an IoT system for real-time monitoring of environmental and safety conditions in mining environments.
- Programmed Arduino Uno to collect sensor data (DHT11, gas, collision, emergency button) and transmit to ESP32 for processing.
- Built backend and edge-to-cloud communication pipelines for live data visualization and automated alerts.
- Debugged firmware–hardware interactions and validated system performance through iterative testing.
- **Tools:** Arduino Uno, ESP32, DHT11, Gas & Collision Sensors, Emergency Button, LCD, Wi-Fi, Edge-to-Cloud Integration, ReactJS

Bachelor's of Science Computer Science

May 2023

University of Missouri-Kansas City

GPA: 3.25

Bachelor's Projects

Graduate Teaching Assistant Job Board

Jan 2023-May 2023

- Developed a web portal for graduate assistants using the MERN stack (MongoDB, Express, React, Node.js).
- Implemented admin dashboards and student workflows with JWT-based authentication for secure access.
- Applied Agile methodologies, leading weekly code reviews and feature planning sessions.
- **Tools:** MongoDB, Express, React, Node.js, JWT, Agile

Energy Company Billing System

Jan 2023-May 2023

- Developed a desktop application to manage customer billing for a simulated energy company.
- Implemented database-backed features for account management, usage tracking, and invoice generation.

- **Tools:** Java, Swing GUI, MS SQL Server 2022, GitHub

Introduction to Artificial Intelligence - Neural Network Project

Jan 2023-May 2023

- Developed neural networks from scratch using NumPy and applied to heart disease prediction
- Built modular architecture with custom layers, achieving binary classification on medical dataset
- Compared multiple model architectures and preprocessing techniques for optimal performance
- **Tools:** Python, NumPy, Keras, TensorFlow, Scikit-learn

EXPERIENCE

Embedded Software Engineer, Orange EV

Oct 2025-Present

- Developed embedded software for heavy-duty electric yard trucks, focusing on seamless hardware-software integration for critical vehicle control systems.
- Led the software integration of a new high-performance traction motor and inverter system for the Husk-e truck model, enhancing powertrain capabilities for heavy-duty port operations.
- Analyzed and optimized existing CAN bus communication networks, improving real-time data transmission reliability and efficiency across vehicle subsystems.
- Utilized Hardware-in-the-Loop (HIL) and Software-in-the-Loop (SIL) simulation environments to validate and debug firmware safety and performance prior to physical deployment.

Graduate Student Research Assistant, University of Missouri-Kansas City

Jun 2023-Jul-2023

- Researched and evaluated deep learning models for image classification in defense related projects
- Preprocessed large image datasets and utilized GPU-based training pipelines for model optimization
- Presented technical findings and contributed to research documentation
- **Tools:** Python, JupyterLab, TensorFlow, PyTorch

Technical Apprentice, Cerner Corporation (Oracle Health)

Aug 2019-Nov 2021

- Developed ReactJS components and NodeJS APIs to support EHR workflows with role-based access control
- Participated in Agile teams, contributing to testing, bug triage, and deployment
- Ensured secure, production-ready releases in a regulated healthcare environment
- **Tools:** GitHub, Jenkins, Jira, REST APIs, JSON, Agile/Scrum